Build book

Big Muff π

MADE	PEDALS	LINES TO BUY	PARTS TO BUY
7 November 2026	1	31	169

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IN THIS PACK

Labels (Avery 5160).pdf
Labels for each part's bag or drawer, 30 to a sheet.

Shopping list.csv
Every part to buy, for a spreadsheet.

Supplier cart.csv
For Mouser's or Digi-Key's BOM upload page.

Stock after these builds.csv
Your stock once this order arrives and these pedals are built. Give it to Star Ground as your stock next time.

Every part comes from the parts lists you gave Star Ground. Check anything marked for checking against its document before you buy.

Shopping list

What to buy for every build in this book, after your stock and any parts that come with a board. “Buy” includes spares where parts are cheap and easy to lose.

Transistors

PART	NEEDED	IN STOCK	BUY	NOTES
BC549C	4	0	**5**	

Diodes

PART	NEEDED	IN STOCK	BUY	NOTES
1N914	4	0	**10**	Swaps to try: 1N4001 (Smooth / Tube-like: Slow reverse recovery (30µs) smears highs).
LED	1	0	**10**	

Potentiometers

PART	NEEDED	IN STOCK	BUY	NOTES
100 kΩ	2	0	**2**	
500 kΩ	1	0	**1**	

Capacitors

PART	NEEDED	IN STOCK	BUY	NOTES
470 pF `471` ceramic	3	0	**8**	Class 1 ceramic (C0G/NP0).
4.7 nF `472` film	1	0	**6**	Box Film.
10 nF `103` film	1	0	**6**	Box Film.
1 µF `105` film	8	0	**9**	Box Film. The document may mean an electrolytic: check it.

Resistors

PART	NEEDED	IN STOCK	BUY	NOTES
100 Ω 	3	0	**10**	1/4 W metal film, 1%.
2.2 kΩ 	1	0	**10**	1/4 W metal film, 1%.
3.3 kΩ 	1	0	**10**	1/4 W metal film, 1%.
10 kΩ 	3	0	**10**	1/4 W metal film, 1%.
27 kΩ 	2	0	**10**	1/4 W metal film, 1%.
39 kΩ 	1	0	**10**	1/4 W metal film, 1%.
47 kΩ 	3	0	**10**	1/4 W metal film, 1%.
100 kΩ 	4	0	**10**	1/4 W metal film, 1%.
470 kΩ 	4	0	**10**	1/4 W metal film, 1%.

Hardware/Misc

PART	NEEDED	IN STOCK	BUY	NOTES
1590B Enclosure	1	0	**1**	
3PDT FOOTSWITCH PCB	1	0	**1**	
3PDT STOMP SWITCH	1	0	**1**	
6.35MM JACK (MONO)	1	0	**1**	
6.35MM JACK (STEREO)	1	0	**1**	
AWG 24 Hook-Up Wire	3	0	**3**	
Bezel LED Holder	1	0	**1**	
DC POWER JACK 2.1MM	1	0	**1**	
Dust Seal Cover	3	0	**3**	
Heat Shrink Tubing	1	0	**1**	
Knob	3	0	**3**	
Rubber Feet (Black)	4	0	**4**	
9V BATTERY CLIP	1	0	**1**	

CHECKLIST

Big Muff π

For one pedal: 45 parts in build order, lowest first, so the board sits flat on the bench as each taller part goes in. Tick each line as it goes in.

01 Resistors

Bend each pair of legs to the holes, solder, then trim.

	QTY	PART	WHERE	WHICH WAY
☐	3	100 Ω	R5, R10, R15	
☐	1	2.2 kΩ 	R21	
☐	3	10 kΩ	R6, R11, R20	
☐	2	27 kΩ 	R16, R17	
☐	1	39 kΩ	R1	
☐	3	47 kΩ 	R4, R9, R14	
☐	4	100 kΩ 	R3, R8, R13, R19	
☐	4	470 kΩ 	R2, R7, R12, R18	

02 Diodes

The band marks the cathode: match it to the stripe on the board.

	QTY	PART	WHERE	WHICH WAY
☐	4	1N914	D1-D4	Band to the stripe

03 Small capacitors

Film and ceramic capacitors fit either way round.

	QTY	PART	WHERE	WHICH WAY
☐	3	470 pF `471` ceramic	C2, C5, C8	
☐	1	4.7 nF `472` film	C11	
☐	1	10 nF `103` film	C10	
☐	8	1 µF `105` film	C1, C3, C4, C6, C7, C9, C12, C13	

04 Transistors

Match each flat face to the outline, and leave a few millimetres of leg.

	QTY	PART	WHERE	WHICH WAY
☐	4	BC549C	Q1-Q4	Flat face to the outline

05 Potentiometers

Fit the board in the enclosure first, so the shafts line up with the holes.

	QTY	PART	WHERE	WHICH WAY
☐	2	100 kΩ	VR1, VR2	
☐	1	500 kΩ	VR3	